Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/03/2018
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Global Transcriptome Profiling to Identify Cellular Stress Mechanisms Responsible for Spaceflight-Induced Antibiotic Resistance		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Merritt Island	**State:**	FL
Zip Code:	32953	**Congressional District:**	3
Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	10/01/2014	**End Date:**	09/30/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 9/30/2020 per NSSC information (Ed., 10/24/2019) NOTE: Extended to 9/30/2019 per NSSC information (Ed., 1/11/2019) NOTE: Extended to 9/30/2018 per NSSC information (Ed., 10/11/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fajardo-Cavazos, Patricia Ph.D. (University of Florida)		
Grant/Contract No.:	NNX14AT38G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Responses of various microorganisms to the spaceflight environment have been examined by numerous studies conducted either in space or on the ground in clinostats. The results clearly show that several microbial behaviors differ in space, but not in a consistent way. The underlying mechanism(s) involved are not explainable by a single model. Exposure to the spaceflight environment (e.g., microgravity, radiation) alters a microbial cell's immediate surroundings, lowering convective mass and heat transfer or reducing mechanical shear forces. Changes in such fundamental physical forces affect the rates at which gases, nutrients, signaling molecules, and waste products are exchanged between microbes and their surroundings. Microbes perceive these alterations as environmental stress (i.e., the "spaceflight syndrome") and mount a complex set of stress responses. One consequence of the spaceflight stress response is alteration in their susceptibility to antibiotics. This could be due to mechanisms such as: production of biofilms, alteration of cell surfaces, up-regulation of drug efflux systems, or an increased rate of mutation in genes encoding drug targets. All of these mechanisms have been well documented as outputs of global microbial stress responses. Of particular interest is oxidative stress, which has been shown both to trigger maladaptive responses in higher organisms including astronauts, and to increase bacterial resistance to antibiotics. Central Hypothesis: Microorganisms subjected to the integrated spaceflight environment invoke a spectrum of stress responses, some leading to alterations in their antibiotic susceptibility. The underlying mechanisms can be identified using transcriptome profiling of model organisms. Specific Aims: We postulate that spaceflight stress responses vary depending on the organism studied because each species of microbe is endowed with its own unique collection of stress response systems. In order to probe the underlying mechanism(s) involved it will be necessary to study spaceflight stress response at the genomic level using model organisms. Specifically we plan to: A. Expose the model bacteria Bacillus subtilis and Staphylococcus epidermidis to: (i) simulated microgravity in the laboratory and (ii) actual spaceflight in 2 parallel Biological Research in Canisters (BRIC) canisters (BRIC-A and BRIC-B) aboard the International Space Station (ISS), with appropriate ground controls. Cells cultivated in BRIC-A will be fixed rapidly with RNA Later and cells in BRIC-B will be frozen in a viable state for return to Earth. B. Total RNA will be extracted from cells in BRIC-A and subjected to transcriptome analysis using Whole Transcriptome Shotgun Sequencing (a.k.a. RNA sequencing or "RNA-Seq") at the University of Florida Interdisciplinary Center for Biotechnology Research (UF-ICBR), to identify the suite of stress responses induced by exposure to spaceflight or simulated microgravity. Cells in BRIC-B will be thawed and immediately assayed for (i) viability, (ii) resistance levels to a battery of antibiotics in the Omnilog system, and (iii) rates of spontaneous mutation to antibiotic resistance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exposure of microorganisms to spaceflight provokes a variety of stress responses, some of which have been postulated to lead to increased antibiotic resistance. The central goal of this proposal is to identify these stress responses, using high-throughput global transcriptome and phenotypic analyses. Understanding how potentially pathogenic microbes might develop resistance to antibiotics in response to stress has significance both for long-term human space exploration and for medical science on Earth.
Task Progress:	[EDITOR's NOTE January 2019--project has now been extended to September 30, 2019; thus, this is not the FINAL report] FINAL REPORTING SUMMARY--JULY 2018: Bacillus subtilis cells were cultivated in 60-mm Petri plates in liquid medium in BRIC-PDFU hardware in two experiments to the International Space Station, designated BRIC-21 and BRIC-23. These experiments flew on SpaceX resupply Cargo Resupply (CRS) missions 6 and 9. Growth of samples was initiated onboard, terminated by freezing, and returned to Earth on the same mission. It was noted that frozen samples from space were in a toroidal (doughnut) shape, different from ground controls which were frozen as disks in the bottom of plates. Frozen samples were recovered and total RNA was extracted. RNA was subjected to ribosomal RNA depletion and RNA-sequencing. Data were processed through a standardized bioinformatics pipeline for analysis. Data showed that several sets of genes were up-regulated in space, most notably those encoding biofilm-associated functions. Ground control samples showed evidence of gene expression related to oxygen limitation, possibly related to the different geometries assumed by the liquid cultures in space vs. ground. ANNUAL REPORTING FOR SEPTEMBER 2016-AUGUST 2017, AS OF AUGUST 2017: Year 3 activities, BRIC-21 mission to the ISS: 1. Numbers of Bacillus subtilis viable cells were found not to be significantly different in Flight (FL) vs. Ground Control (GC) samples. 2. We have completed antibiotic resistance screening of B. subtilis FL and GC samples to a battery of 72 different antibiotics and growth-inhibitory compounds using Omnilog Phenotype Microarray (PM) assays. We found no significant difference in the resistance of FL vs. GC samples to 63 of these compounds. FL samples were more resistant than GC samples to 3 antibiotics, and more sensitive to 6 antibiotics. Of these nine compounds, we further tested FL vs. GC samples for resistance levels by standard broth-dilution assay and found no significant difference in the resistance levels of FL vs. GC samples. These results have been submitted for publication. 3. We have completed isolation of rifampicin-resistant RifR mutants from the BRIC-21 Flight (FL) and Ground Control (GC) experiments. We found that FL and GC samples mutated to RifR with essentially equal frequencies. We determined the nucleotide sequences of these mutations in the Bacillus subtilis rpoB gene encoding the beta subunit of RNA polymerase. We found that FL and GC samples exhibited dramatic differences in the spectrum of mutations within rpoB. These results are being prepared for publication and will be submitted before the end of Year 3. 4. We have completed extraction of total RNA from FL vs. GC samples, have performed RNA-seq, and have received the unprocessed raw data. We are currently analyzing these 12 data sets using standard bioinformatics software. We plan to request a 1-year no-cost extension in order to finish the data analysis and publish the results.

Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Morrison M, Fajardo-Cavazos P, Nicholson WL. "Transcriptome and antibiotic resistance profiling of Bacillus subtilis cells exposed to the human spaceflight environment on the BRIC-21 mission to the International Space Station." Poster at the Microbiology Graduate Student Symposium, Amelia Island, FL, August 19-20, 2016. Proceedings book. Abstracts of the Microbiology Graduate Student Symposium, Amelia Island, FL, August 19-20, 2016. , Aug-2016
Abstracts for Journals and Proceedings	Morrison M, Fajardo-Cavazos P, Nicholson WL "Exposure to spaceflight conditions affects the transcriptome and antibiotic resistance profiles of Bacillus subtilis cells on the BRIC-21 Mission to the ISS." Florida Branch of the American Society for Microbiology, Key Biscayne, FL, October 14-16, 2016. Program and Abstracts of the Florida Branch of the American Society for Microbiology, Key Biscayne, FL, October 14-16, 2016. Abstract #26, p. 31. , Oct-2016
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Articles in Peer-reviewed Journals	Zeigler DR, Nicholson WL. "Experimental evolution of Bacillus subtilis." Environ Microbiol. 2017 Sep;19(9):3415-22. Review. Epub 2017 Jul 26. https://doi.org/10.1111/1462-2920.13831 ; PubMed PMID: 28631363 , Sep-2017
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Articles in Peer-reviewed Journals	Fajardo-Cavazos P, Nicholson WL. "Establishing standard protocols for bacterial culture in Biological Research in Canister (BRIC) hardware." Gravit Space Res. 2016 Dec;4(2):58-69. http://gravitationalandspacebiology.org/index.php/journal/article/view/744 , Dec-2016
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